Air Quality Python Hypothesis and Forecasting Report by Vamshi Aitharaju

Description of my Code

For Python Analysis I have chosen Hypothesis Testing and Forecasting as my data set was completely based on Season and Years. So, to do this as I don't have handful experience in python, I took AI help and did not completely relied on it, I have only used AI to tell me what kind of Libraries I need to use such as Pandas for reading my excel data first, NumPy for math calculations, SciPy for Statistical testing and Hypothesis, Stats models for forecasting, finally Matplotlib is for Visualizations and Plots, so I learnt and used libraries which are only required for my analysis part in python. I have Opted with the Exponential Smoothing approach in python too as excel did took the same, so I have opted for the same approach for better accurate results. I did two separate Python scripts for my analysis.

The first one, air quality hypothesis, which is mainly for checking seasonal and pollutant differences using hypothesis testing. I opened the sheet name Hypothesis Testing from my Excel file and used only the main columns I need like Winter PM2.5, Summer PM2.5, Winter NO_2, Summer NO_2, Annual PM2.5, and Annual NO_2. Before doing any test, I cleaned my data by removing all empty cells using the dropna() command. After that I calculated the average for each column and used the scipy.stats package for testing. For checking the difference between winter and summer, I used ttest_ind() which is independent test. For yearly comparison between Annual PM2.5 and Annual NO_2, I used ttest_rel() which is paired test. The code gave the average, difference, p-value, and small message showing if it is significant or not. The second file is air quality forecasting, which I have used for predicting air quality for next four years that is from 2024 to 2027. I used Exponential Smoothing model from statsmodels. For that I worked with my Excel file Air Quality Prepared which I made before by filtering my EDA file by myself as if I do same in coding is it bit difficult and lengthy process. I read it using pd.read_excel(), changed the date column using pd.to_datetime(), and sorted the values. Then the model predicted data from 2024 to 2027 for five sheets, Annual NO_2, Annual PM2.5, Winter NO_2, Summer NO_2, and Combined PM2.5(Combined because in first part I got P values greater than 0.05). Finally, I used matplotlib for charts, plt.plot() for lines and plt.fill_between() for confidence range, which shows 95 percent which I did same in excel.

How Python adds value to my decision making

Doing this in Python help me to check if both Excel and Python results are same or not. As I already did forecast in Excel, I want to prove it again in Python to see if any changes happen. This way I can know that my excel forecast is correct, because Python gives more accurate values. So if both results are same then it means my analysis is on the right track. If not I can see where it is changing and why. By doing both excel and python, I can trust my results with more accuracy and also learn how forecasting behave in Python. It makes my project stronger because I checked in two tools and got clear understanding and yes, for my project result were almost same with little changes In values, so when I research about this difference with TA Marley she explained that it is because of python and excel has **different rounding system** which I saw on internet too.

Approaches and methods used for Hypothesis Testing

For my hypothesis part I did same what I done in Excel but now in Python. I used the same test that I did before in EDA because I already worked a lot with that data. First I checked same pollutant in different seasons like Winter PM2.5 vs Summer PM2.5 and Winter NO_2 vs Summer NO_2. After Professor Kristin told me to try, I also compared different pollutants in same season like Annual PM2.5 vs Annual NO_2. I wanted to do the same thing again in Python and see if the results are same or any small change happen I just want to see if result will change or it stay same like Excel. So I asked AI to give me small code for forecasting and to make chart same like my Excel one. It told me to use Exponential Smoothing model from statsmodels because it is simple and give smooth trend line. I opened my Excel file Air_Quality_Prepared.xlsx in Python by using pandas. I used pd.read_excel() to open and pd.to_datetime() to fix my date column. Then I sort my data and used command ExponentialSmoothing(df["Value"], trend="add", seasonal=None) Then I fitted model with .fit(smoothing_level=0.5, smoothing_trend=0.25, optimized=False) and forecasted till 2027 using .forecast(steps=4). I also asked AI how to make chart same like Excel, and it told me to use plt.plot() for line and plt.fill_between() to show 95% confidence area. When I run it, it show both old and future data in same chart. So I can compare if both follow same trend. This part help me to understand that Python also giving same direction like Excel. Forecast for NO_2 is going down and PM2.5 is also slowly down. So here by I got to know that my results are same in excel and python.

Summary of What My Code Predicts

In Python I used Date and Value for each pollutant like PM2.5 and NO_2 from 2009 till 2023. I used this data to see how values will change in coming years. The code give me forecast for 2024, 2025, 2026 and 2027. In Excel I already saw that NO_2 is going down every year and PM2.5 is also going down but very slow. When I did same in Python, I got almost same result. The chart also look same but only small change in few values because Python calculate more correct with decimals. The pattern and direction are same. I checked both Excel and Python side by side to see if there is any big change. Both showing same trend. NO_2 is more high in winter and less in summer. PM2.5 is almost same all time. Only small difference in numbers came because of rounding. So both are matching and correct. In both charts, I got blue line for history and orange for forecast. In Python I also got small shade area which show confidence level. It show that air quality is getting better slowly because pollution is reducing little by little every year. When I see both pollutants, I can say they are not related because their pattern not same. NO_2 change more with season, PM2.5 not much. NO_2 is higher overall, PM2.5 stay low and steady. This show both behave different and need to see separate. From the output, I can clearly say that **NO_2 is still high in winter and low in summer. PM2.5 stay same in both**. But both showing small improvement every year, may be due government policies and after Covid the pollution started going down. So Python also giving same forecast like Excel, Forecasting is verified in excel and python.

Hereby I, Vamshi Aitharaju, declare that following work is done by me and is ready for review.